

Alkanes, Alkenes, and Alkynes

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Contents

1	Alkenes	2
1.1	Alkene Synthesis	5
1.2	Alkene Addition Reactions	8
1.2.1	Catalytic Hydrogenation	8
1.2.2	Electrophilic Additions	9
1.2.3	Hydroboration-Oxidations	14
1.2.4	Electrophilic Carbene Additions	16
1.2.5	Electrophile Oxidation	18
1.2.6	Radical Hydrobromination	22
2	Alkynes	23
2.1	Alkyne Synthesis	25
2.2	Alkyne Reactions	26
2.2.1	Reductions	26
2.2.2	Electrophilic Additions (Markovnikov)	28

1 Alkenes

Core Concept 1.1: Alkene Orbital Structure

The orbital structure of an alkene double bond is a combination of a σ bond and a π bond. Compared to the alkane single bond length of roughly 1.5 Å, the alkene double bond is shorter at approximately 1.3 Å.

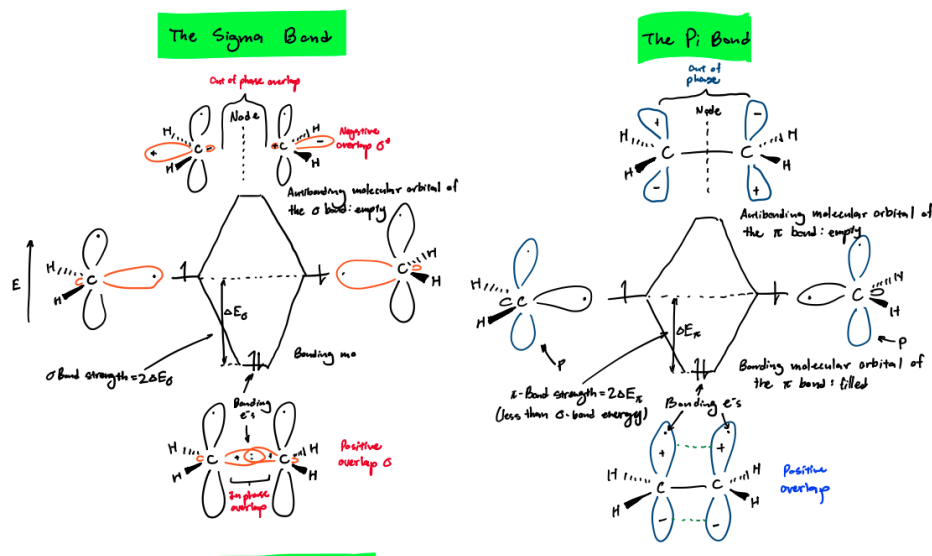


Figure 1: caption

Core Concept 1.2: Alkene Bond Strength

The π bond of an alkene is relatively weaker than its partner σ bond because its corresponding antibonding π^* orbital sits closer to the bonding π orbital than the antibonding σ^* orbital with the σ bonding orbital. As a result, breaking an alkene π bond takes roughly 65 kcal mol^{-1} whereas the σ bond $108 \text{ kcal mol}^{-1}$.

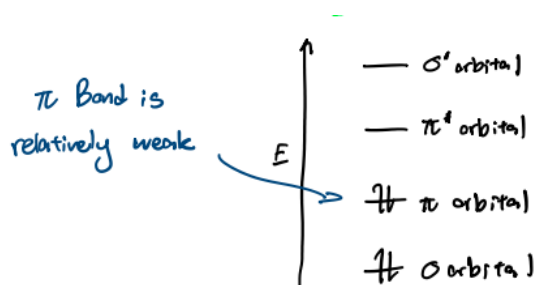


Figure 2: caption

When compared with alkanes, the typical alkene σ C-C bond strength is roughly 88 kcal mol^{-1} . Notice that the alkene σ bond is stronger than the alkane σ bond because the alkene σ bond uses sp^2 hybridized orbitals compared to the sp^3 orbitals of alkanes, as shown in FIGURE REFERENCE. This higher s orbital characteristic—specifically 33% vs 25%, respectively—means greater bond strength e⁻s are pulled in closer relative to p orbitals.

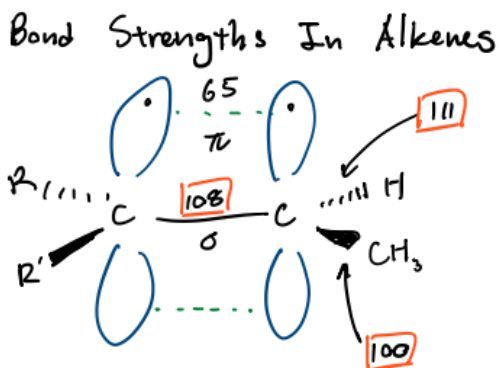


Figure 3: this may have to be made again

Thus in the context of the pK_a values of substituent hydrogens of alkenes and alkanes, the alkenes show lower values compared to that of alkanes.

<chemfig of ethene and ethane with pKa>

Therefore, in principle, the following reaction between methyllithium and an alkene will reach a product favoring equilibrium because the alkene is more acidic (*able to release a proton better*) than the alkane.

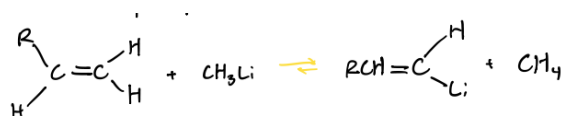


Figure 4: caption

However, in practice, the reaction in FIGURE REFERENCE will face regio- and stereoselectivity issues because the other substituent hydrogens can equally or more favorably dissociate compared to the one shown. In practice, chemists utilize the formation of **organometallics** (Core Concept 2.3) to overcome these problems.

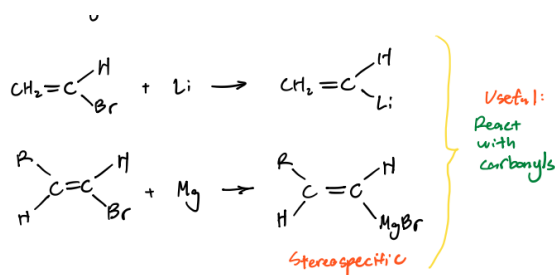


Figure 5: caption

Core Concept 1.3: Relative Stability of Alkenes

Consider the **heat of hydrogenation** ($\Delta H_{\text{hydrogenation}}$) of following butene isomers to butane.

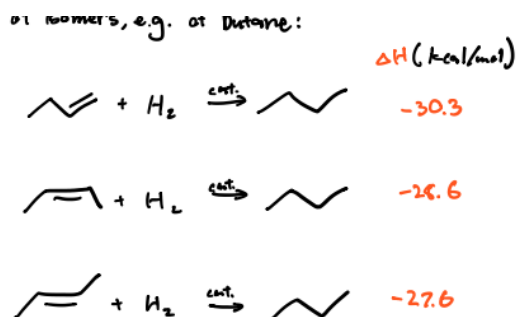


Figure 6: caption

The relative greater stability of internal over terminal double bonds and trans over cis substituents arise from two reasons.

- (a) **Hyperconjugation** allows the temporary charge imbalance arising from the transitionary state to be better distributed by neighboring molecular orbitals in the internal double bond butene compared to the terminal counterpart.

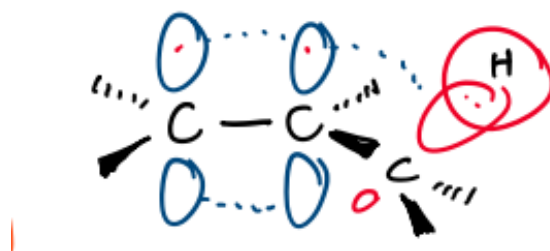


Figure 7: caption

- (b) **Steric hindrance** between the cis methyl groups increase the relative instability of the butene compared to the trans counterpart.

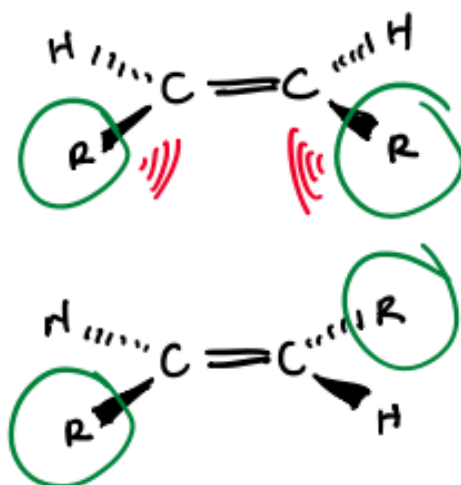


Figure 8: caption

Thus the general order of alkene stability acts as follows.

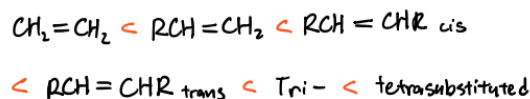


Figure 9: caption

1.1 Alkene Synthesis

Core Concept 1.4: Elimination & Saytzev Rule

Synthesizing alkenes from alkanes through E2 elimination requires the consideration of several factors.

(a) Base Size

Consider an E2 reaction **with a small base** on the following RX alkane.

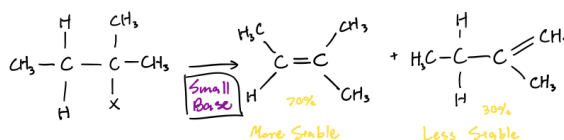


Figure 10: caption

Notice the favoring of the internal double bond alkene compared to the terminal version because the advantages of hyperconjugation win over the minimal effects of steric hinderance due ot the use of a small base. In contrast, an E2 reaction **using large base** (BELOW FIGURE REFERENCE) favors the terminal version because steric hinderance exerts a comparatively larger influence on transition state stability enough to override to hyperconjugation favorability of the internal double bond product.

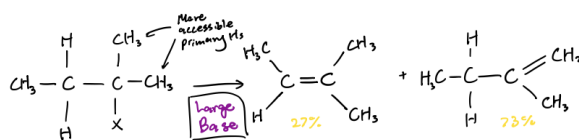


Figure 11: caption

This phenomenon is termed the **Saytzev Rule**, which states that non-bulky bases lead to more stable internal double bonds. The corresponding energy diagrams of the two reactions are shown below to visually depict the energy levels of the reaction pariticipants throughotu the reaction.

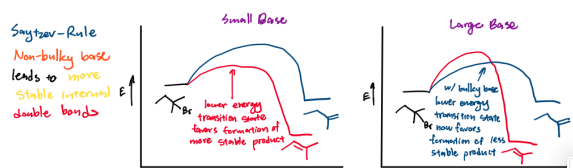


Figure 12: caption

(b) Stereoselectivity

The presence of **multiple stereoisomers**—specifically among the formed internal double bond alkenes when utilizing E2 reactions with small bases—is something a chemist must also consider. Although the E2 reaction is stereoselective, favoring the more stable trans isomers over the sterically hindered cis isomers, the relationship is not absolute (*some cis isomers do form*).

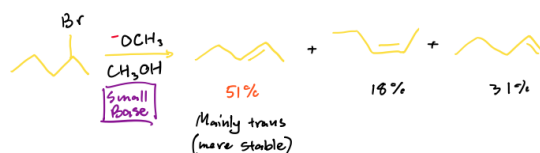


Figure 13

(c) Other Factors

Other factors such as preferable precursor conformations, namely rotamers, play a role as well.

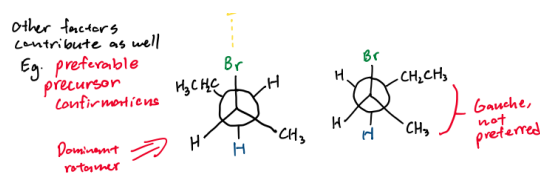


Figure 14

Keep in mind that elimination is also **stereospecific** as shown in the following figure.

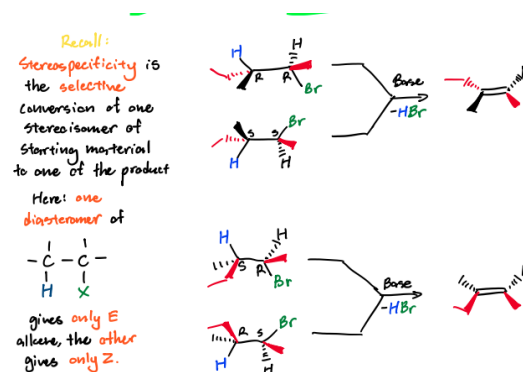


Figure 15

Core Concept 1.5: Alkenes from ROH By Dehydration

Alkenes can also be synthesized from exposing alcohols to acids and subsequently removing the resulting H_2O byproducts through dehydration.

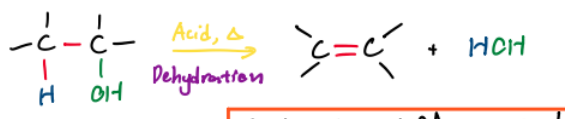


Figure 16

The mechanism behind the reaction—using sulfuric acid and ethanol as an example—is as follows.

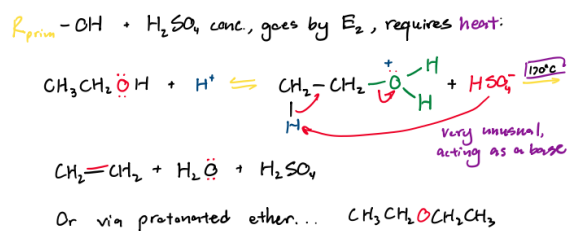


Figure 17

Due to the effects of steric hinderance and hyperconjugation, the usage of E_1 or E_2 and resulting stereoisomer product ratio differs across $\text{R}_{\text{primary}}$, $\text{R}_{\text{secondary}}$, and $\text{R}_{\text{tertiary}}$.

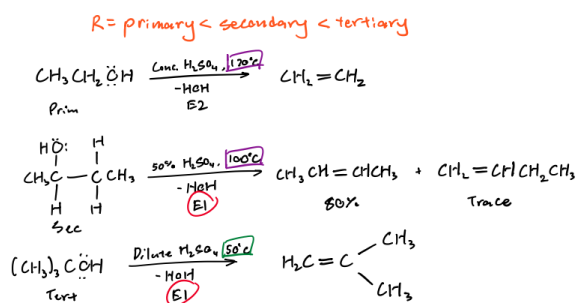


Figure 18

A huge limitation to this is the formation of a **carbocation transition state**, which opens the door to **unintended rearrangement pathways** such as **hydride** (Core Concept 3.1) or **alkyl shifts** (Core Concept 3.3) if possible. As a result, this alkene synthesis pathway has limited utility in practice.

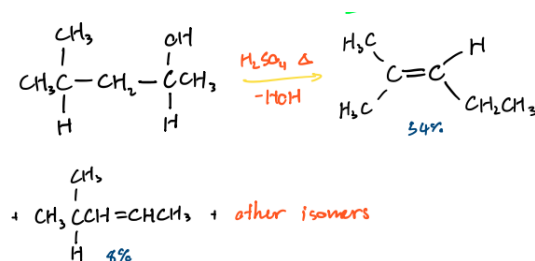


Figure 19

1.2 Alkene Addition Reactions

Core Concept 1.6: Alkene Addition Reactions

Alkenes are the fundamental building blocks for organic synthesis and the preparation of polymer materials. Their defining feature is the **unsaturated π bond**, which reactions through **addition reactions**, wherein the π bond is broken and a diatomic or compound molecule is added across the relevant carbon atoms of the former alkene.

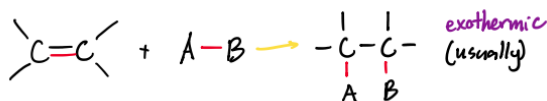


Figure 20

Generally speaking, alkene addition reactions are $\Delta H^\circ < 0$, however since the nature of the reaction is a combination reaction wherein two reactant molecules form a single product molecule, $\Delta S^\circ < 0$, making it entropically unfavorable. In most reactions, the enthalpic favorability overrides the magnitude of entropic unfavorability. However, in some reactions or at high temperatures, the ΔG° turns positive, making these addition reactions thermodynamically unfavorable.

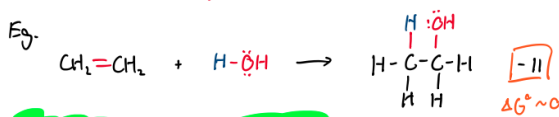


Figure 21

1.2.1 Catalytic Hydrogenation

Core Concept 1.7: Catalytic Hydrogenation

Catalytic hydrogenation refers to the addition of H₂ to an alkene, effectively removing the alkene double bond and converting the target molecule into an alkane if the double bond in question was the only one. This process alone would require enormous amounts of pressure and extreme temperatures to occur, however **catalysts**—such as Pd/C or PtO₂—enable a mechanism with lower E_a that chemists frequently use.

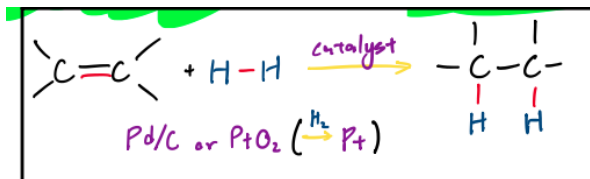


Figure 22

It's also important to note that the hydrogens are **stereospecifically** added to the **same side (syn)** of the target double bond. As a result, the outputted product mixture will be **racemic**.

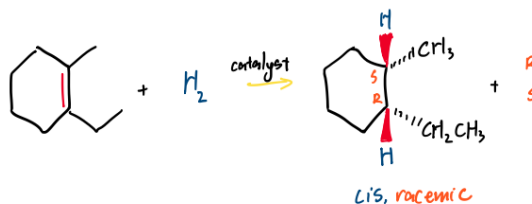


Figure 23

1.2.2 Electrophilic Additions

Core Concept 1.8: Electrophilic Additions

Because the π bond is e^- rich, it readily reacts with a polar reagent $A^{\delta+}-B^{\delta-}$, where the partially positive end $A^{\delta+}$ is drawn to the e^- rich double bond.

In the scenario where $A^{\delta+}$ is a H^+ , the reaction that ensues is simply the reverse of an E1.

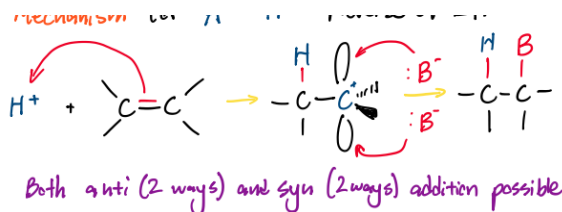


Figure 24: indicate anti and syn stereoisomers in caption

Core Concept 1.9: Hydrohalogenation

Hydrohalogenation is a type of electrophilic addition wherein a **HX is added** to the alkene double bond where X is a halogen.

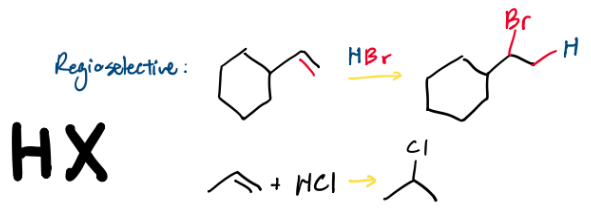


Figure 25

Notice that the H^+ adds to the **less substituted** carbon because it results in a more substituted carbocation in the other carbon, favoring tertiary > secondary > primary carbocation formation. This phenomenon is called **Markovnikov's Rule** and the following energy diagram for the different stereospecific pathways depicts the stability of the different transition states nicely.

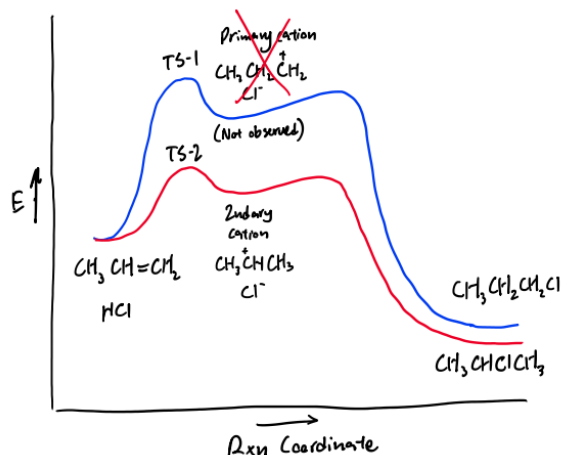


Figure 26

Core Concept 1.10: Markovnikov Hydration

Markovnikov hydration refers to the conversion of an alkene to an alcohol through the **addition of H_2O** in the presence of an acid catalyst—the reverse of the alkene synthesis reaction covered in Core Concept 1.5.

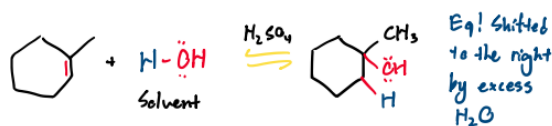


Figure 27

The underlying mechanism to the reaction is shown in the following. Notice how the H^+ is regenerated by the end of the reaction.

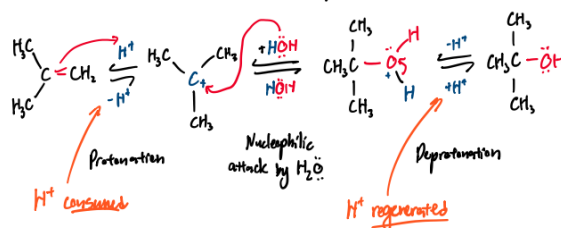


Figure 28

There are, however, bound to be a mixture of alkenes, transient intermediaries, and alcohols because Markovnikov hydration is at dynamic equilibrium with the pathway previously covered in Core Concept 1.5. Because the temporary presence of a carbocation introduces hydride shifts, catalytic H^+ will turn a single alkene species into a slurry of stereoisomeric alkenes, with ratios determined by relative thermodynamic stabilities primarily based on steric hindrance and hyperconjugation.

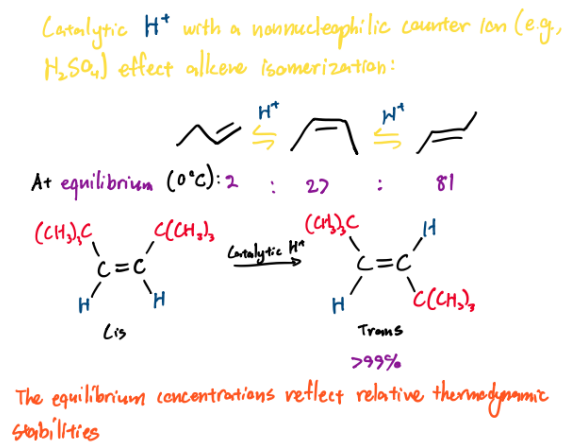


Figure 29

The full mechanism of equilibrium between an alcohol and its corresponding alkene products is shown in the following chemical equation using butanol/butene as an example.

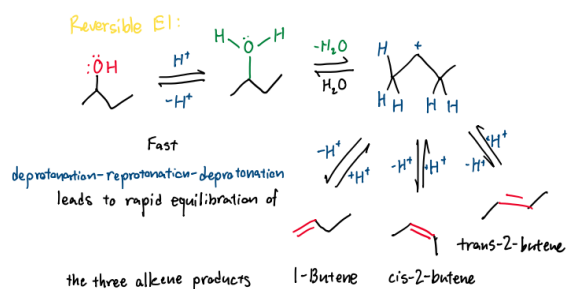


Figure 30: caption includes: leads to the rapid equilibrium of three alkene products

Core Concept 1.11: Halogenation

Halogenation is the addition of a diatomic halogen molecule X_2 to an alkene double bond. In contrast to the addition of HX (Core Concept 1.9)—which is inherently polarized due to the electronegativity differences— X_2 is not polarized inherently, but rather **gets polarized during its approach to the e^- rich alkene double bond**.

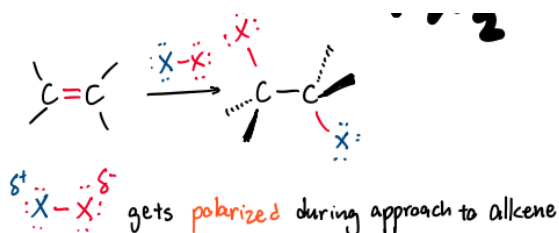


Figure 31

The underlying mechanism is shown in the following figure.

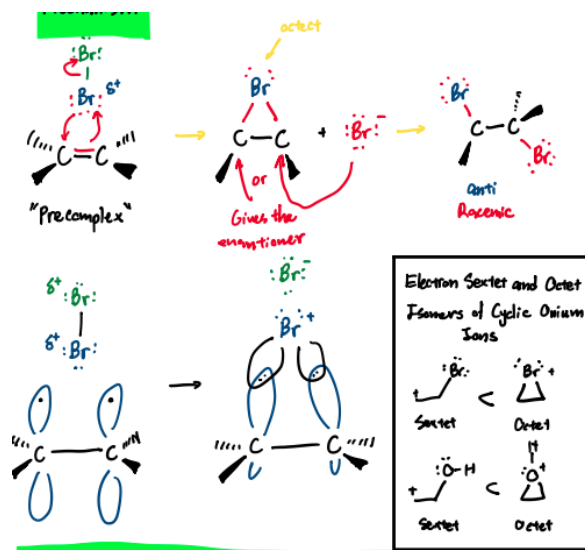


Figure 32

Notice the formation of a halonium ring because the alternative carbocation state will want to distribute the burden of its positive charge. Because the second attack to the bridged halonium exclusively occurs to the backside of the first halogen attack, halogenation is **stereospecifically anti (not syn)**. Likewise, if the second attack has thermodynamically equal possibility of attack either backside carbon—meaning that the initial alkene was perfectly **symmetrical**—the ensuing products will be an **racemic mixture**.

If the initial alkene is **asymmetrical**, the halonium ion intermediate is actually highly distorted, wherein the bond between the halogen and more substituted carbon is **longer**. Likewise, as the halogen pulls more e^- density towards itself the more substituted carbon is able to possess a greater δ^+ due to hyperconjugation. As a result, the halonium ring transition state **already has a high degree of carbocation** behavior and thus mimics more of a $\text{S}_\text{N}1$ attack characteristic although the backside attack is $\text{S}_\text{N}2$ by definition because of its exclusively anti result. Consequently, the backside nucleophilic attack occurs at the more substituted carbon, an interaction wherein the impact of charge is greater than steric hindrance on regioselectivity.

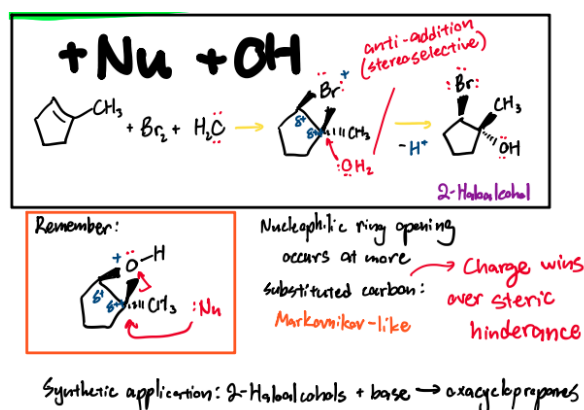


Figure 33

On a different note, the intermediary halonium ring ions can also be intercepted by other nucleophiles such as HOH (FIG REFERENCE ABOVE) or ROH (FIG REFERENCE BELOW).

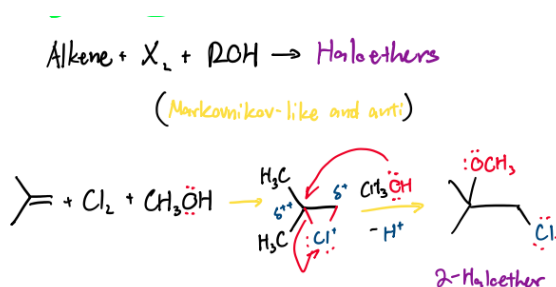


Figure 34

Core Concept 1.12: Oxymercuration–Demercuration

Oxymercuration–demercuration is effectively the same as the **Markovnikov hydration** (Core Concept 1.10) process wherein a H_2O is added to remove an alkene double bond. However, oxymercuration–demercuration **avoids carbocation** formation, which prevents the unintended rearranged product alcohols and produces a purer product mix.

The process utilizes mercury (II) acetate and sodium borohydride in a two step chemical process to achieve its H_2O addition.

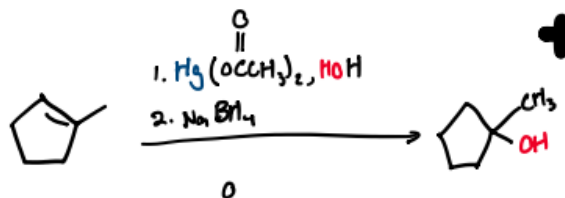


Figure 35

<figure of simple H_2O equation>

(1) Oxymercuration

Similar to the **alkene halogenation** (Core Concept 1.11) process, the large mercury atom forms a bridged, unevenly distributed mercurinium ion rather than a full carbocation, forcing the subsequent H_2O attack to strictly occur at the more substituted carbon.

(2) Demercuration

Afterwards, the NaBH_4 acts as a reducing agent that cleaves the C-Hg bond and replaces it with a simple hydrogen atom.

Likewise, the usage of ROH allows the addition of a $-\text{OR}$ group just as how a $-\text{OH}$ was added by H_2O .

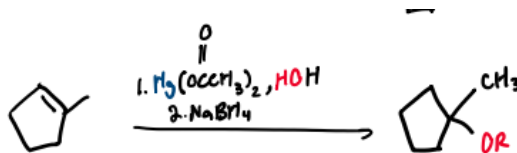


Figure 36

When compared to Markovnikov hydration (Core Concept 1.10), oxymercuration–demercuration is the preferred method used by chemists in order to convert an alkene to an alcohol due to the purity of the resulting product mix.

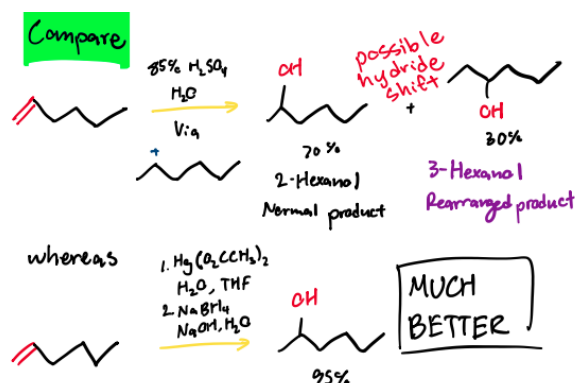


Figure 37

1.2.3 Hydroboration-Oxidations

Core Concept 1.13: Hydroboration-Oxidations

Hydroboration-oxidation is a method chemists use to add H_2O to a double bond of an alkene at the **less substituted carbon of the double bond** (remember that the Markovnikov hydration and oxymercuration–demercuration H_2O additions occur at the more substituted carbon). This process utilizes BH_3 to regioselectively bond to 3 alkenes and subsequently oxidate and cleave the resulting BR_3 to produce 3 ROH.

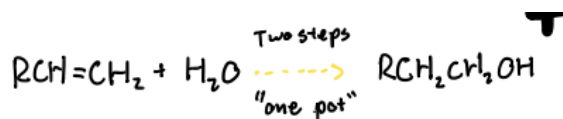


Figure 38

(1) Hydroboration

When mixed with BH_3 , three alkenes bond to BH_3 with the **less substituted carbon paired to the boron**, occurring due to two major factors.

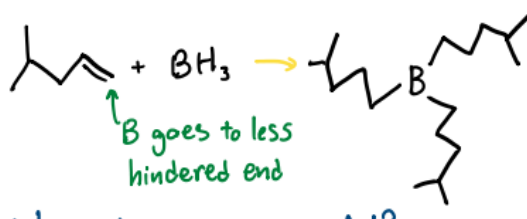


Figure 39

- **Transition State Stabilization**

When the e^- rich π bond of the alkene reaches out to attack the empty unhybridized p orbital of

the boron, the two carbon atoms of the double bond do not share the burden equally. The carbon that is **not** forming the C-B bond begins to lose its e^- density, creating a partial positive charge δ^+ during its transition state. Consequently, if the less substituted carbon forms the C-B bond, the more substituted carbon can better stabilize this charge through hyperconjugation, allowing for a more stable transition state relative to the alternative C-B bond formation.

<need to make figure>

• Bond Polarization

Another subtle factor that plays a role is the inherent electronegativity difference between boron (2.0) and hydrogen (2.1), creating a polarized bond where boron is slightly positive (δ^+) and hydrogen is slightly negative (δ^-). As a result, the slightly positive boron more favorably pairs with the e^- rich attack from the less substituted alkene carbon and the slightly negative hydrogen more favorably pairs with the subsequently partial positive non-boron bonding alkene carbon.

<need to make figure>

Another important characteristic of hydroboration is the **syn** stereospecificity of the resulting alkylborane. The B and H that are attacked by an alkene π orbital are physically attached to one another and added to the double bond simultaneously, forcing them to attach to the same face of the alkene.

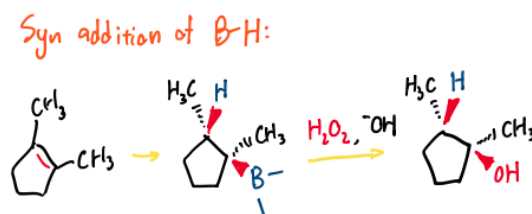


Figure 40

(2) Oxidation

The resulting alkylborane R_3B is then oxidized by a solution of H_2O_2 and ^-OH which yields trialkyl borate $((RO)_3B)$ through the following alkyl shift mechanism.



Figure 41

Finally, H_2O hydrolyzes the $(RO)_3B$ with the help of a base to form the intended 3 ROH molecules with a borate ion as a byproduct.

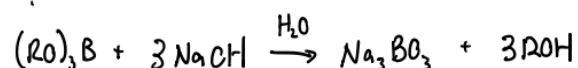


Figure 42

Often this entire process is called **anti-Markovnikov hydration** because it adds a H_2O molecule to the alkene

but does so in a manner where the -OH group ends up on the less substituted π bond carbon rather than the more substituted one.

1.2.4 Electrophilic Carbene Additions

Core Concept 1.14: Carbene Synthesis

Carbenes (:CR_2) are sp^2 hybridized 6e^- species with a lone pair and empty p orbital—the parent of which is methylene (:CH_2). They can be thought of as being derived by **deprotonating carbocations**.

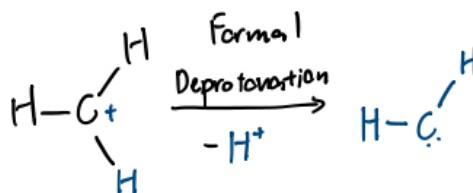


Figure 43

Because they are e^- deficient, carbenes are extremely unstable, short lived intermediates that act as powerful **electrophiles**. The following are the three primary pathways chemists use to generate these intermediates in the laboratory.

(a) Methylene from diazomethane (CH_2N_2)

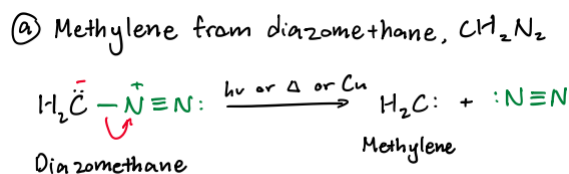


Figure 44

Mind that this reaction is practically irreversible because the resulting N_2 is incredibly stable, driving the reaction forward to cleanly leave behind highly reactive methylene.

(b) Dichlorocarbene from chloroform

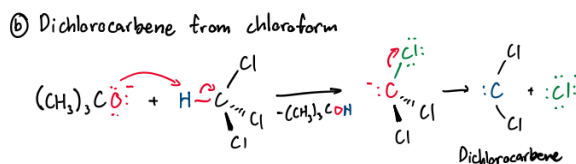


Figure 45

The above reaction utilizes α -elimination (distinct from the standard β -eliminations E1 or E2 normally used to make alkenes) through the strong base $(\text{CH}_3)_3\text{CO}^-$ and the departure of a highly electronegative chlorine atom.

(c) Simmons-Smith reagent

As mentioned previously, free methylene is so wildly reactive that it can be incredibly dangerous to work with. The Simmons-Smith reaction provides a much safer, controlled workaround.

③ Simmons-Smith reagent

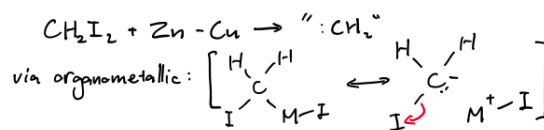


Figure 46

The treatment of diiodomethane with a zinc-copper couple forms an organometallic species that behaves like a hybrid between a stable molecule and a methylene like state.

Core Concept 1.15: Electrophilic Carbene Additions

Being the naturally reactive species that they are, carbenes can be added to alkenes in a **syn stereospecific manner** through the following general mechanism, forming carbon ring as a product.

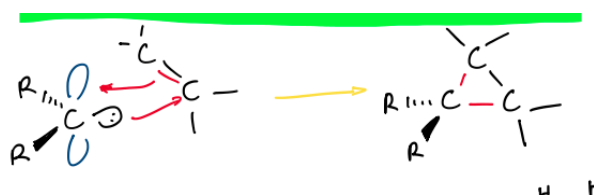


Figure 47

In practice, since carbene species are transient, this carbene addition must occur alongside the carbene synthesis processes (Core Concept 1.14).

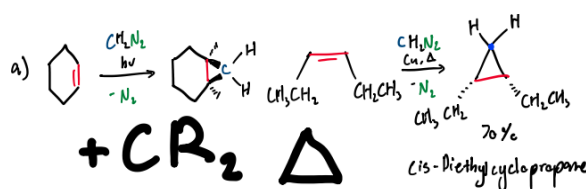
(a) Methylene from diazomethane (CH_2N_2)

Figure 48

(b) Dichlorocarbene from chloroform

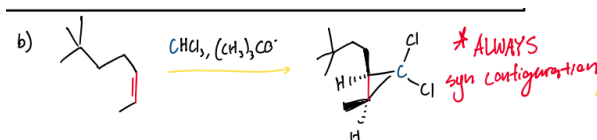


Figure 49

(c) Simmons-Smith reagent

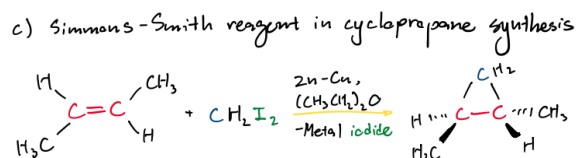


Figure 50

1.2.5 Electrophile Oxidation

Core Concept 1.16: Peroxycarboxylic Acids

Peroxycarboxylic acids are a species of acids with the following general structure.

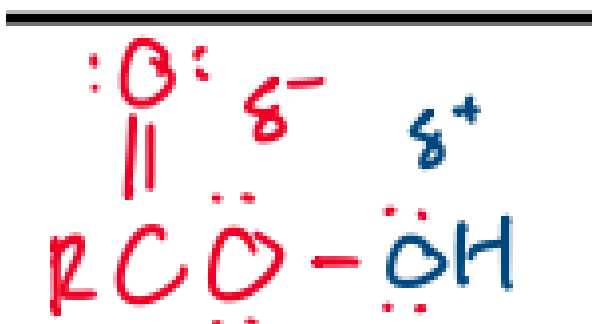


Figure 51: add the commonly used examples as well to the figure so it is 4 molecules side by side

When alkenes are treated with these compounds, the reaction results in a **syn addition of an oxygen** that forms a cyclic ether product as outlined below.

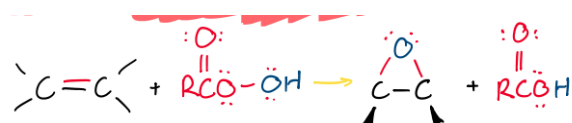


Figure 52: outline with red and blue coloring to represent charges and dashes and wedges to represent the stereospecificity

The underlying **concerted** mechanism looks like the following.

Mechanism: concerted

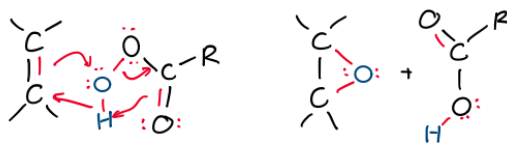


Figure 53

It's also important to note that the **rates of oxacyclopropanation increases with alkyl substitution** because increased hyperconjugation allows to a greater degree of e^- donation from the electrophilic alkene π orbital.

Rates of oxacyclopropanation increases w/ alkyl substitution: hyperconjugation = e^- donation

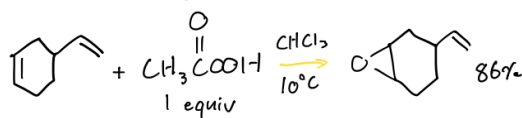


Figure 54

In practice, peroxycarboxylic acid addition usually serves as a preparatory step for subsequent modifications. Two common applications are shown below.

(a) Double Nucleophile Syn-Addition

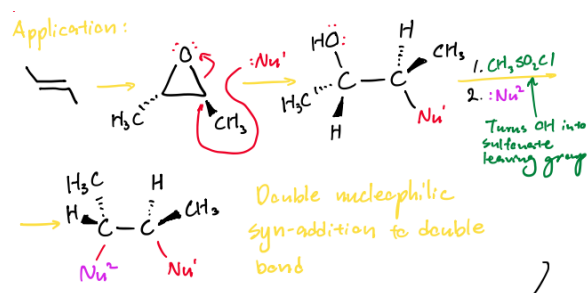


Figure 55

(b) Anti Stereospecific Dihydroxylation

Sequence 1. RCO_3H , 2. H^+ , H_2O or OH^- , H_2O

constitutes an **anti-dihydroxylation** of alkenes

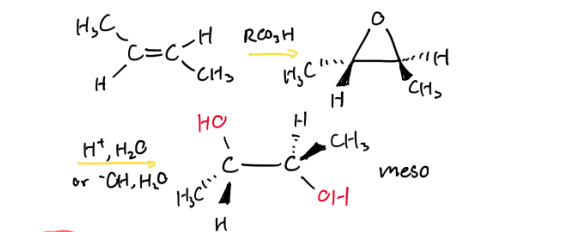


Figure 56

Core Concept 1.17: OsO_4 Syn-Dihydroxylation

The treatment of alkenes with OsO_4 and a subsequent reduction through an agent like H_2S results in syn-dihydroxylation along an alkene double bond.

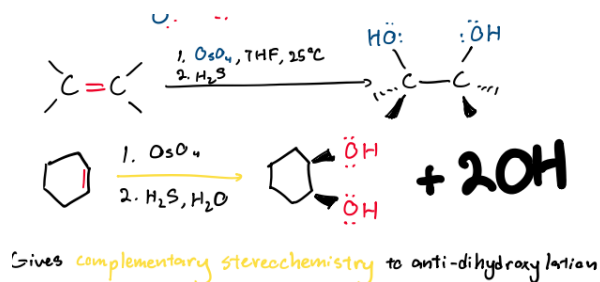


Figure 57

The underlying mechanism proceeds as follows, creating osmate ester intermediary product before the H_2S step reduction.

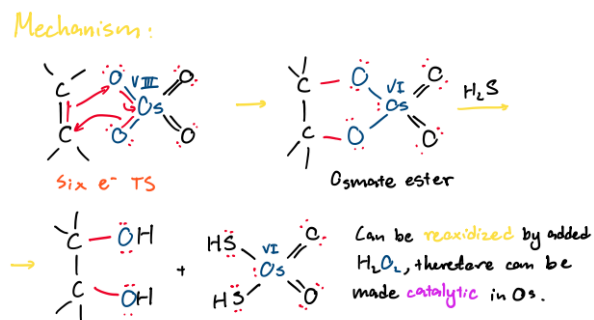


Figure 58

The consequently used Os compound can be reoxidized by adding H_2O_2 , allowing the Os compound to be **catalytic** (not consumed by the reaction until H_2O_2 depletes). This is helpful because Os is expensive.

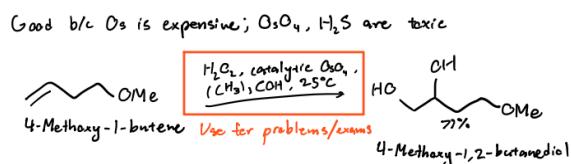


Figure 59

Core Concept 1.18: Ozonolysis

Ozonolysis utilizes ozone (O_3)—a highly unstable and rapidly decaying species—and a reducing agent to perform a complete oxidative cleavage of an alkene double bond.



Figure 60

Because ozone is such a transient molecular species, it isn't available enough naturally chemists to work with it. To prevent this shortage, chemists put O_2 through an **arc discharge**—continuous, highly intense electrical breakdown of a gas—to generate fresh stream of 2–4% O_3 gas safely in a lab.

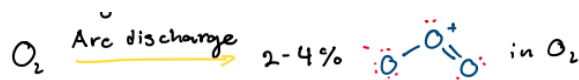


Figure 61

The following are three of the most common reducing agents used by chemists.

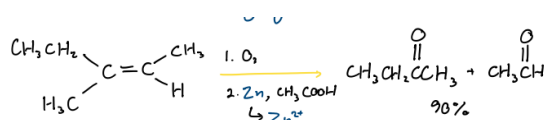
(a) Zinc and Acid (Zn / CH_3COOH)

Figure 62

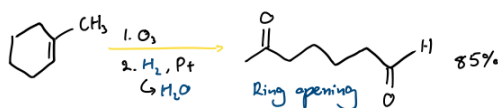
(b) Catalytic Hydrogenation (H_2 / Pt)

Figure 63

(c) Dimethyl Sulfide (DMS; $(CH_3)_2S$)

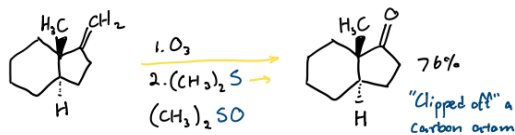


Figure 64

1.2.6 Radical Hydrobromination

Core Concept 1.19: Radical Hydrobromination

Radical hydrobromination refers to the **addition of an HBr** molecule to an alkene double bond with the assistance of $RO-OR$ in an **anti-Markovnikov manner**, meaning that the Br is attached to the **less substituted carbon**. Recall that regular **hydrohalogenation** (Core Concept 1.9) follows Markovnikov regiochemistry where the halogen is attached to the more substituted carbon.

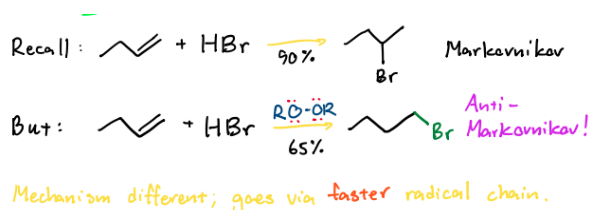


Figure 65

The underlying mechanism is shown in the following figure, started by peroxide radical initiators (the $O-O$ bond is weak at around 39 kcal mol^{-1})

(a) Initiation Steps

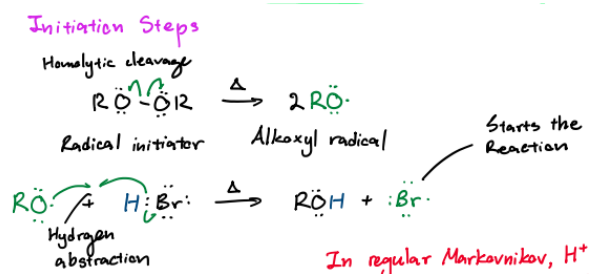


Figure 66

(b) Propagation Steps

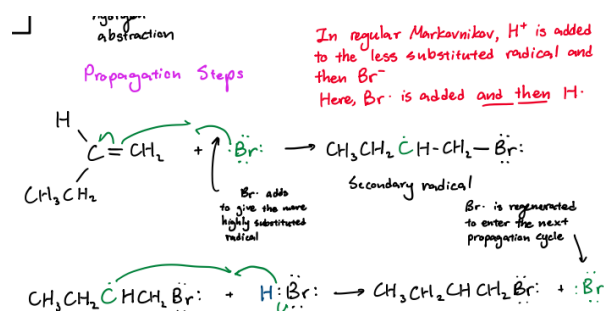


Figure 67: separate propagation I and II in the first

Notice that the $Br\cdot$ is added to the less substituted carbon, resulting in the resulting carbon radical to be better hyperconjugated than the alternative binding arrangement due to being more substituted.

This radical hydrobromination does not work for HCl or HI because one of the propagation steps are **endothermic** due the unique electronegative properties of Cl and I . Hence the radical mechanism is too slow and the ionic mechanism in the Markovnikov manner (*attaching to more substituted carbon*) is favored. Br , on the other hand, sits at the thermodynamic sweet spot between Cl and I . RSH anti-markovnikov addition works in the same manner as Br , sitting in that same sweet spot.

2 Alkynes

Core Concept 2.1: Alkyne Orbital Structure

An alkyne triple bond is composed of two perpendicular π bonds and a conjoined sp hybrid σ bond.

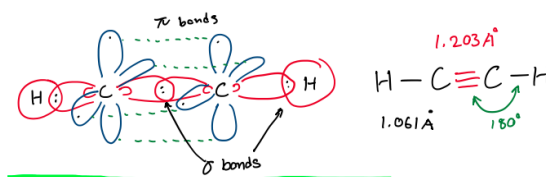
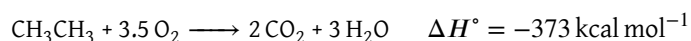
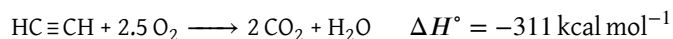


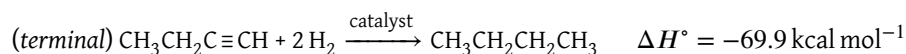
Figure 68

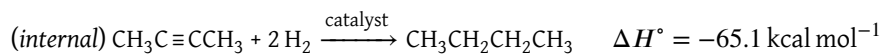
Core Concept 2.2: Triple Bond Strength

Ethyne releases a massive $-311 \text{ kcal mol}^{-1}$ of heat when combusted. Compared to ethane (CH_3CH_3)—which releases -373 kcal/mol —ethyne releases a proportionally greater amount of energy on a per carbon basis because you don't have to spend extra energy to break the 4 additional $C-H$ bonds in ethane.



Due to the higher level of strain alkyne π bonds exist under, the full hydrogenation of the two alkyne π bonds releases more energy than that of two alkene π bonds (*which would be roughly $-60 \text{ kcal mol}^{-1}$*)





Notice the differences in net energy release due to the greater stability of the internal triple bond over the terminal version because of hyperconjugation.

Core Concept 2.3: Relative Acidity of Alkynes

In general hydrocarbons (alkanes and alkenes), the C-H bond is considered completely non-acidic. However, terminal alkyne C-H bonds defy this rule when treated with a sufficiently strong base, wherein the terminal H^+ can be plucked off.

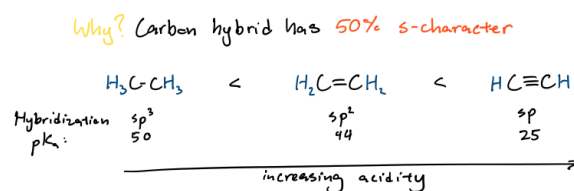
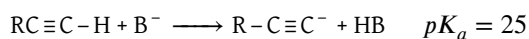


Figure 69

The above acidic trend is explained by the greater s orbital characteristics of the corresponding orbital hybridization. Since an s-orbital is spherical and sits much closer to the positively charged carbon nucleus than a directional, elongated p-orbital, the lone pair of electrons left behind after deprotonation is held very close to the positive nucleus. This electrostatic proximity stabilizes the negative charge immensely, making the corresponding alkyne conjugate base comparatively more stable.

Using this alkyne acidity, chemists are able to execute the following example applications

(a) Organolithium Reagents



Figure 70

(b) Sodium Amide (NaNH_2)

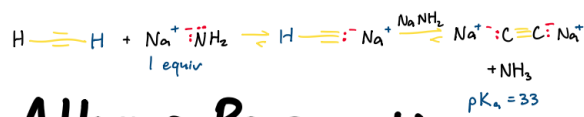


Figure 71

2.1 Alkyne Synthesis

Core Concept 2.4: Elimination E2 From Dihaloalkanes

Performing E2 elimination twice on dihaloalkanes with adjacent halogen substituents allows for alkyne formation.

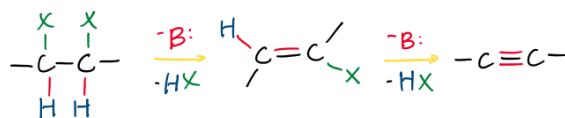


Figure 72

Here is a specific example utilizing NaNH_2 as the base, which has to go through an extra aqueous work up step due the acidity of terminal alkynes (Core Concept 2.3) in the presence of sodium amide NaNH_2 .

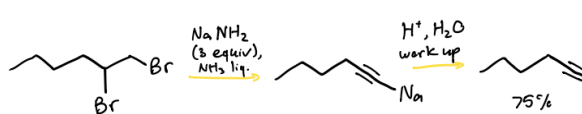


Figure 73

Core Concept 2.5: Alkylation of Alkynyl Metals

Terminal alkynes can be converted into powerful nucleophiles via deprotonation with strong bases, enabling C-C bond formation through substitution pathways that follow strict $\text{S}_{\text{N}}2$ rules.

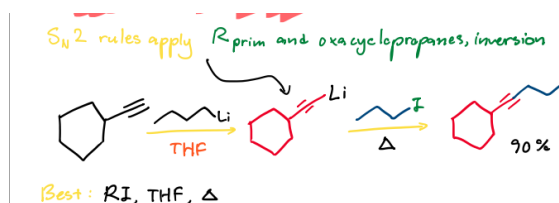


Figure 74

The synthetic utility of alkynyl metals relies on R_{prim} halides or oxacyclopropanes (epoxides) as electrophilic substrates to minimize competing E2 elimination side-reactions. When utilizing unsymmetrical epoxides, the acetylide nucleophile regioselectively attacks the less sterically hindered carbon atom.

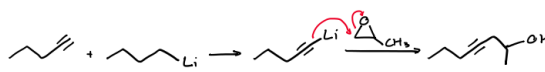


Figure 75

The following are various examples using this mechanism.

(a) Carbonyl Addition with Grignard

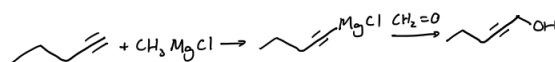


Figure 76

(b) Epoxide Opening with Lithium Acetylide

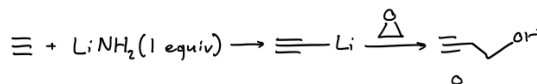


Figure 77

(c) Double Grignard Addition of Acetaldehyde

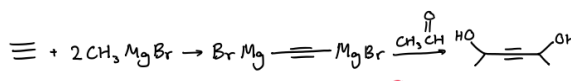


Figure 78

2.2 Alkyne Reactions

2.2.1 Reductions

Core Concept 2.6: Complete Hydrogenation

The complete hydrogenation of an alkyne triple bond (addition of 2H_2) progresses similarly alkene hydrogenation (Core Concept 1.7), utilizing a catalyst like Pt to help lower the E_a of the reaction. The resulting product is an alkane.

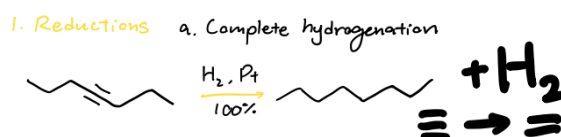


Figure 79

Core Concept 2.7: Partial Hydrogenation

Partial hydrogenation refers to the addition of only one H_2 wherein only one of the two π bonds are removed, resulting in either cis or trans product from a syn and anti addition, respectively.

• Lindler's Catalyst (Syn Addition)

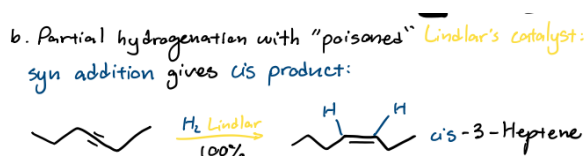


Figure 80

A syn addition and subsequent cis product requires the usage of a "poisoned" **Lindlar's catalyst**.

• **Sodium Reduction (Anti Addition)**

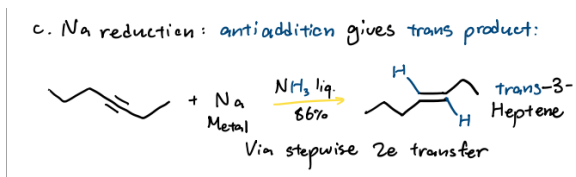


Figure 81

The underlying mechanism for the above reduction through sodium is shown below.

(1) e^- Transfer

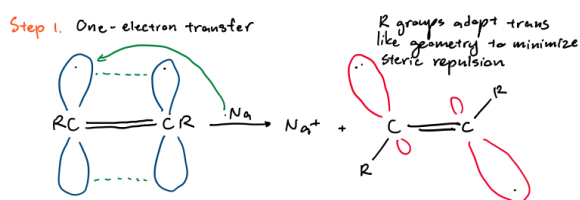


Figure 82

(2) First Protonation

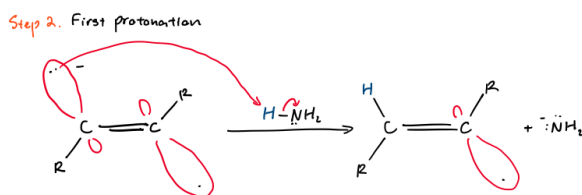


Figure 83

(3) Second e^- Transfer

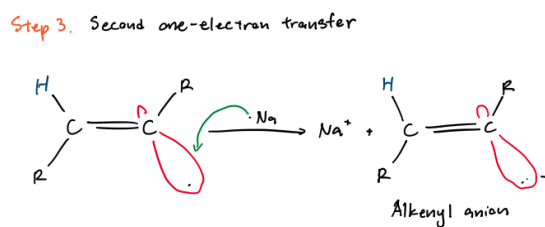


Figure 84

(4) Second Protonation

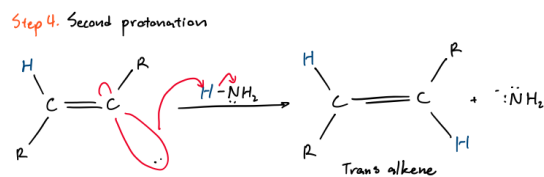


Figure 85

2.2.2 Electrophilic Additions (Markovnikov)

Core Concept 2.8: HX Anti Addition

The **addition of HX** to an alkyne mix converts the triple bond to a **geminal** (same side) dihaloalkane.

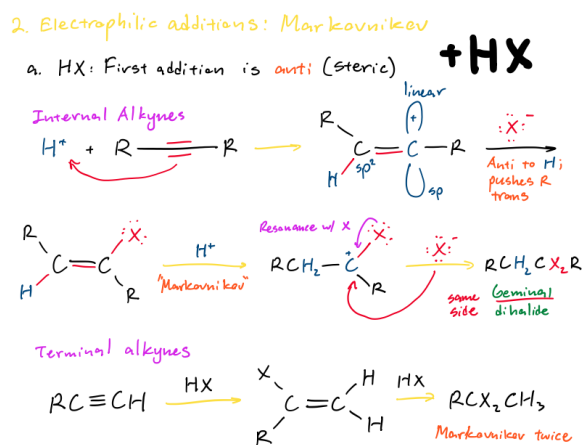


Figure 86

It's important to notice that the first HX addition is always executed in the anti configuration because of steric hinderance.

Additionally, although FIGURE REFERENCE categorizes this HX addition process into internal and terminal alkyne categories, it's much simpler to understand that the ensuing transition states of the reaction favor carbocation formation on the more substituted carbon (or Markovnikov's rule). Because of this tendency, the second HX addition will bind the second H^+ to the same first H^+ receiving carbon because the alternative carbon is better substituted by the presence of a halogen substituent from the first HX addition.

Here are some real reactions, showcasing alkyene HX anti additions.

(a) Symmetrical Internal Alkyne



Figure 87

(b) Asymmetrical Internal Alkyne

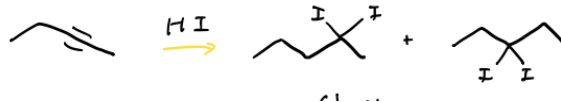


Figure 88

Here because the degree of hyperconjugation between the bonded R groups do not have a large difference, mixtures ensure although the PRODUCT NAME most likely is more favored.

(c) **Terminal Alkyne**

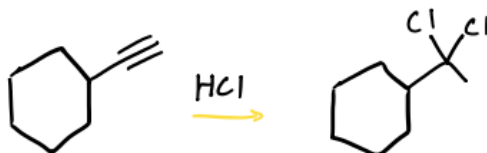


Figure 89

(d) **Synthetic Application (Vicinal to Geminal)**

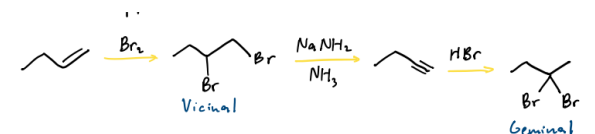


Figure 90

The first step of the chemical reaction of above is alkene halogenation, covered in Core Concept 1.11.

Core Concept 2.9: X₂ Anti Addition

Alkyne halogenation **adds X₂** to an alkyne triple bond, resulting in the addition of four halogens (two to each triple bond carbon) via a transient ring state. The mechanism for these additions an anti arrangement and is effectively alkene halogenation (Core Concept 1.11) occurring twice in a row.

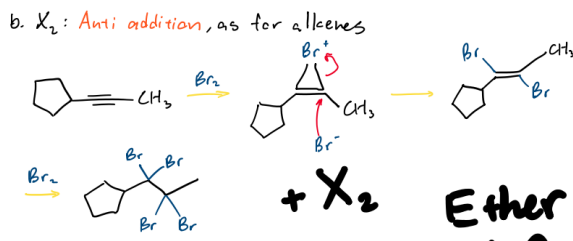


Figure 91

Core Concept 2.10: Mercury Catalyzed Markovnikov Hydration

The treatment of an alkyne with catalytic HgSO_4 (previously seen in Core Concept 1.12) and H_2O leads to an unstable enol intermediate in a Markovnikov manner. The subsequent **tautomerization** of this enol with catalytic H^+ or OH^- leads to the formation of a ketone, regardless of whether the alkyne triple bond is terminal or internal (because the OH group is attached to the more substituted carbon.).

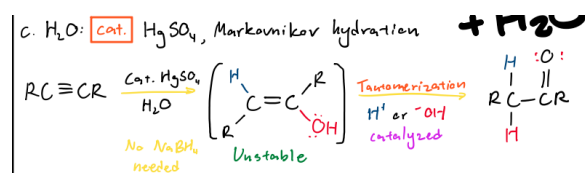


Figure 92

The underlying mechanism of **tautomerization**—both H^+ and OH^- pathways—is shown below.

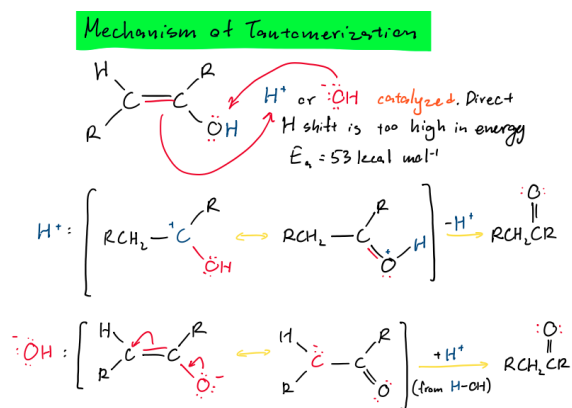


Figure 93

Here are also some practical use cases in the laboratory.

(a) Asymmetrical Internal Alkyne

Examples.

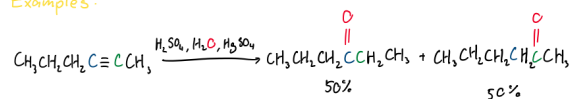


Figure 94

(b) Terminal Alkyne

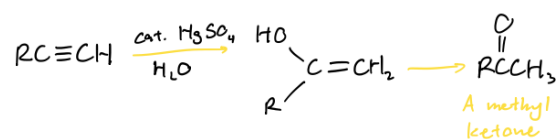


Figure 95

Core Concept 2.11: Radical Hydrobromination

Radical hydrobromination in alkynes is the **single addition of HBr** through the formation of radical species with the assistance of ROOR. The reaction follows the same mechanism covered in the alkene version of this electrophilic addition, covered in Core Concept 1.19, however leads to the formation of product mixes due to the possible rotation along the resultingly formed double bond.

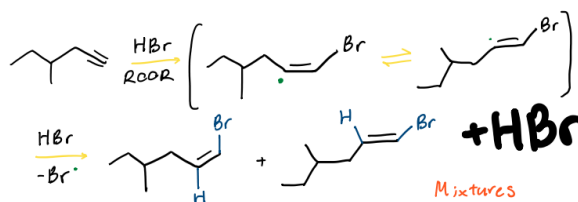


Figure 96

Unlike the additions we've seen in HX anti addition (Core Concept 2.8) and X_2 anti addition (Core Concept 2.9), this reaction does not necessarily undergo a second iteration to convert the resulting alkene further into an alkane form. This is because the mechanism requires the presence of an e^- dense π bond, which the resulting bromine-attached alkene does not satisfy because bromine is highly electronegative, pulling e^- density away from the double bond (*a clean alkene does not experience this*). As a result, because the starting alkyne is significantly more reactive than the bromoalkene product, the radicals will preferentially consume all the remaining alkyne before even attempting to touch the bromoalkene intermediate.

If one exposes alkynes to an excess of HBr and peroxides, however, a second radical addition can occur, which follows the same Markovnikov stability rules to form a **geminal dihaloalkane**, the same product of HX anti addition (Core Concept 2.8), making the extra peroxide use unnecessary if the original intention was to form a geminal dihaloalkane.

Core Concept 2.12: Hydroboration-Oxidation

Alkyne **hydroboration-oxidation** utilizes a R_2BH (R is a bulky group such as cyclohexyl) to ensure the hydroboration of only one π bond. Recall that alkene hydroboration-oxidation (Core Concept 1.13) utilized simply BH_3 because there is only one π bond to begin with. The mechanism follows the same process as the alkene version of this reaction (Core Concept 1.13) and effectively accomplishes an anti-Markovnikov hydration.

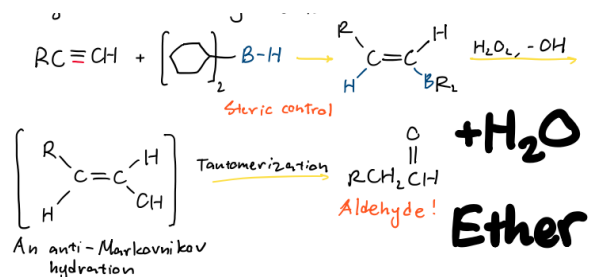


Figure 97: exclude the taut step here

The resulting alkenol can subsequently be put through previously covered processes like tautomerization to be converted into an aldehyde.

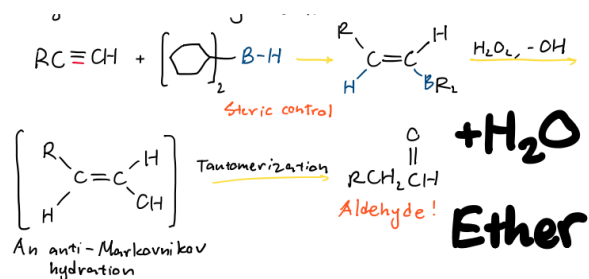


Figure 98: only the taut step here

The following is a direct comparison with **mercury catalyzed Markovnikov hydration** (Core Concept 2.10), which has a similar but distinct outcome resulting in a ketone rather than an aldehyde.

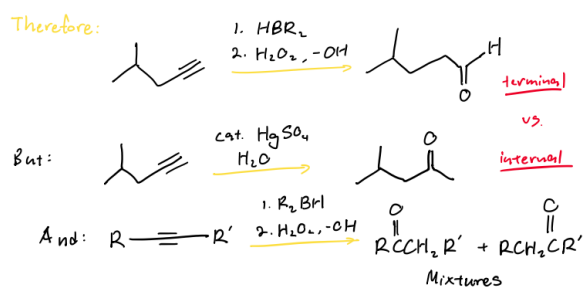


Figure 99

In an internal alkyne scenario, mixtures ensue when the degrees of substitution in the triple bond carbons don't have a large difference (although the slightly less substituted side will be favored in the context of final carbonyl positioning).